

Project Summary: Microgrid Design for a Town

This project focuses on the design and simulation of a microgrid system using a combination of conventional (diesel generator) and renewable (solar PV and battery storage) energy sources. The system is intended to improve grid reliability, reduce dependency on fossil fuels, and maintain system stability under various operating conditions.

Key Objectives:

- Assess overall power system performance
- Increase generator power factor
- Integrate renewable energy sources and evaluate their effects
- Simulate grid-connected and islanded operation modes

Technologies & Tools:

- RSCAD by RTDS for simulation
- Diesel Generator (125 kVA)
- PV System (300 kW), BESS (200 kW)
- Variable load (300 kW to 1 MW)

Highlights:

- Achieved stable frequency (49.83–50.54Hz) in grid-connected mode
- Renewable penetration reached ~80%
- Identified challenges in islanded mode due to frequency instability
- Proposed BESS for fast frequency response in future improvements

Visual Aids:

The following figures illustrate the simulated network, key components, and performance metrics.

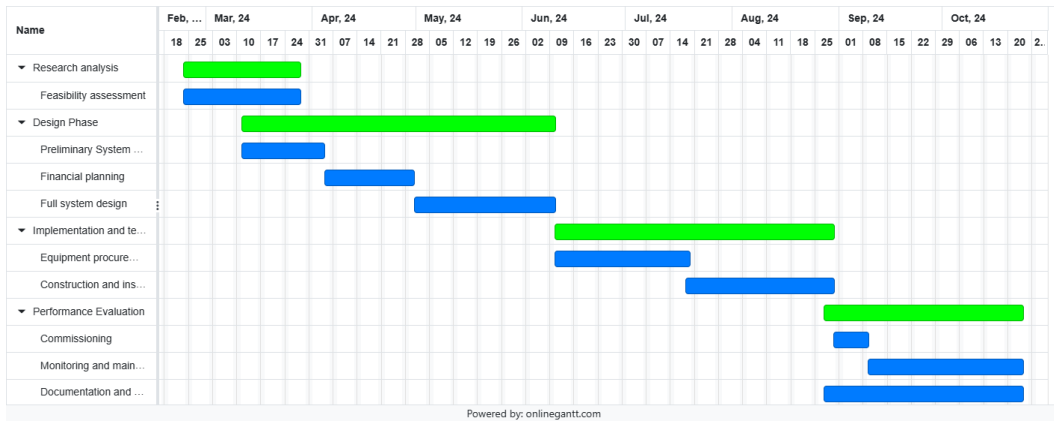


Figure 1: Gantt chart for the project timeline.

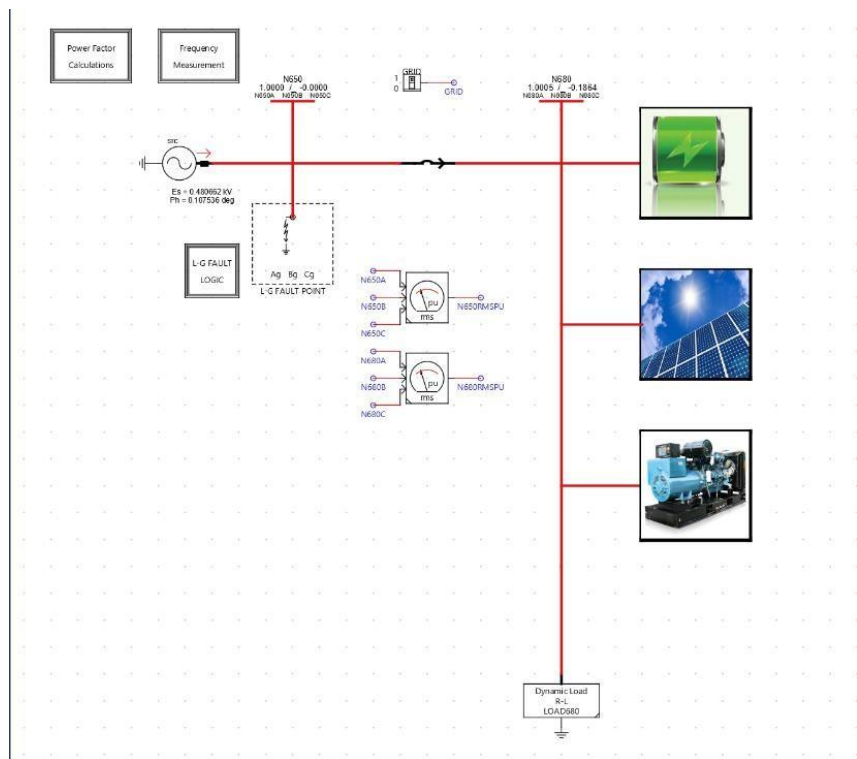


Figure 2: Modified Microgrid example case from RSCAD by RTDS

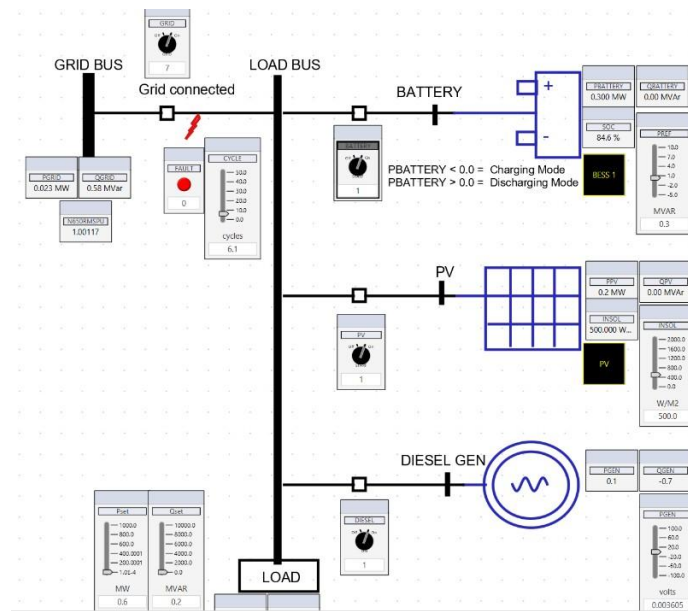


Figure 3: Runtime simulation when Grid-connected and all DERs are ON

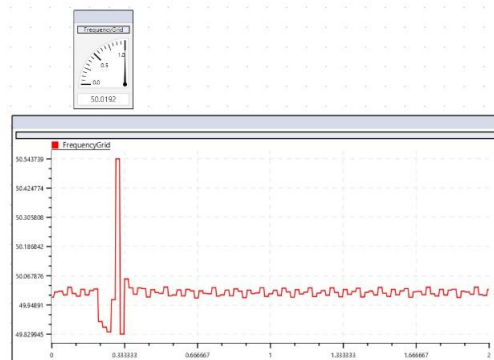


Figure 4 : Frequency of the grid when Grid-connected and all DERs are ON

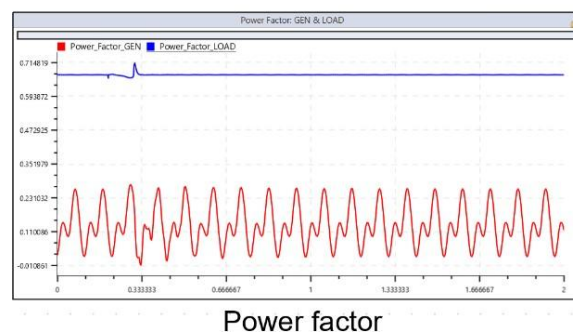


Figure 6 : Power factor for generator and the grid when Grid-connected and all DERs are ON

For Scenario 1: Grid-connected mode when all DERs are connected

Parameter name	Benchmark parameter	Simulated value	Assessment
Frequency	50 $\pm$ 2% (49Hz – 51Hz)	50 Hz	Meet criteria
Load power factor	0.9	0.67	Does not meet criteria
Generators power factor	0.85 - 0.95	Generator not connected	Can not measure
Synchronization time	Instantly	Instantly	Meet criteria
Power matching	Must be balanced	Balanced	Meet criteria
Renewable penetration	50-80%	$\approx$ 80%	Expected value

Table 1: Results comparison from scenario 1

For Scenario 2: Islanded mode

Parameter name	Benchmark parameter	Simulated value	Assessment
Frequency	50 $\pm$ 2% (49Hz – 51Hz)	59 Hz	Does not meet criteria
Load power factor	0.9	0.57	Does not meet criteria
Generators power factor	0.85 - 0.95	0.9	Meet criteria
Synchronization time	200ms	111ms	Meet criteria
Power matching	Must be balanced	Balanced	Meet criteria
Renewable penetration	50-80%	100%	Expected value

Table 2: Results comparison from scenario 2

Conclusion:

The project demonstrates the viability and stability of a hybrid microgrid in grid-connected mode. Further enhancement is needed for standalone operation, particularly with respect to frequency control and power factor correction.